

Engineering
ENGINEER 2PX3
Communication and Social Impact
Winter 2024

INSTRUCTOR INFORMATION

Dr. Maryam Aramesh, Ph.D., P.Eng., prof2px3@mcmaster.ca, JHE 309
Office Hours: by appointment.

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Dr. Scott Chen, Ph.D., prof2px3@mcmaster.ca, ITB A316
Office Hours: by appointment.

Mr. Cameron Churchill, prof2px3@mcmaster.ca, JHE-316/C
Office Hours: by appointment.

Dr. Kim Jones, Ph.D., prof2px3@mcmaster.ca
Office Hours: by appointment.

Dr. Vincent Maccio, Ph.D., prof2px3@mcmaster.ca, ITB 219
Office Hours: by appointment.

Dr. André Phillion, Ph.D., P.Eng., prof2px3@mcmaster.ca, ETB 107
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Dr. John Preston, Ph.D., P.Eng., prof2px3@mcmaster.ca, JHE A214/K
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ADDITIONAL STAFF

Mr. Basam Yassa
Instructional Assistant
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Mr. Jake Wilson
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Ms. Jasmine Sun-Hu
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COURSE INFORMATION

Lectures:

C01: Monday and Wednesday at 11:30 am – 12:20 pm
C02: Monday and Wednesday at 8:30 am – 9:20 am
C03: Tuesday and Thursday at 11:30 am – 12:20 am

Design Studio:

L01: Friday at 2:30 pm – 4:20 pm
L02: Thursday at 8:30 am – 10:20 am
L03: Tuesday at 1:30 pm – 3:20 pm
L04: Thursday at 3:30 pm – 5:20 pm
L05: Wednesday at 12:30 pm – 2:20 pm
L06: Tuesday at 3:30 pm – 5:20 pm

L07: Friday at 4:30 pm – 6:20 pm

L08: Thursday at 11:30 am – 1:20 pm

Tutorial: Asynchronous Online, 1 hour per week

Course Dates: 01/08/2024 - 04/10/2024

Units: 3.0

Course Delivery Mode: In-Person. In-person attendance is required for this course.

Course Description: A multidisciplinary integrated learning course focused on communication and the engineering design cycle. Formal and informal, verbal and written communication skills are fostered through project-based design sprints.

IMPORTANT LINKS

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)

COURSE LEARNING OUTCOMES

1. Perform different steps of the engineering design cycle,
2. Effectively communicate their design process and decisions,
3. Apply principles to demonstrate respectful and productive teamwork, and
4. Articulate what they have learned about the design process and team dynamics.

COURSE LEARNING GOALS

Effective communication skills are essential for success in the engineering profession. In a field driven by innovation and collaboration, engineers must convey complex ideas and technical information with clarity and precision to a range of audiences. Written communication skills enable them to articulate their thoughts in formal and informal documentation, ensuring seamless knowledge transfer and efficient project management. Similarly, strong oral communication skills empower engineers to present, collaborate, persuade, and inform a broad audience (e.g., stakeholders, colleagues, clients, supervisors, etc.). Clear and concise communication fosters understanding, promotes teamwork, and builds trust, ultimately enabling engineers to deliver innovative solutions that drive progress in our interconnected world. This course aims to develop technical communication skills through a series of multidisciplinary design projects.

GRADUATE ATTRIBUTES

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national quality standard and cover essential educational requirements. Graduate Attributes are qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

4.1 Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions. (Learning Outcome 1)
4.4 Justifies and reflects on design decisions, giving consideration to limitations, assumptions, constraints and other relevant factors. (Learning Outcome 1)
6.2 Manages interpersonal relationships, taking leadership responsibilities as needed. (Learning Outcome 3)
7.2 Composes an effective written document for the intended audience. (Learning Outcome 2)
7.3 Composes and delivers an effective oral presentation for the intended audience. (Learning Outcome 2)
8.1 Describes the duty of a Professional Engineer to the public, client, employer, and the profession. (Learning Outcome 4)
9.1 Evaluates the environmental impact of engineering activities, identifies uncertainties in decisions, and promotes sustainable design. (Learning Outcome 2)
9.2 Evaluates the social impact of engineering activities, including health, safety, legal, cultural, and other relevant factors, and identifies uncertainties in decisions. (Learning Outcome 2)
10.2 Applies the principles of equity and universal design to ensure equitable treatment of all stakeholders. (Learning Outcomes 2, 4)
12.1 Critically assesses one's own educational needs and opportunities for growth. (Learning Outcome 4)

COURSE SCHEDULE		
Week	Lecture 1	Lecture 2
1	Intro	What is Design?
2	Evaluation Framework	Evaluation Framework (cont.) + Justification
3	Written Communication I	Written Communication II
4	Sprint 2 Simulation Intro + Brainstorming I	Group Dynamics I
5	Group Dynamics II	Stakeholder Identification
6	Oral Communication I	Oral Communication II
7	Reading Week	
8	Problem Statement/Requirements (Activity) + Human-centered Design	Project Management (Gantt Chart, Critical Path)
9	Engineering Judgement	Oral Communication III
10	Written Communication III	Reflection on Equitable Design
11	Improving Engineering Design	Persuasion/ Fallacies
12	Engineering Case Studies	Engineering Case Studies
13	Presentations	Presentations
14	Presentations	Presentations

Note: This structure represents a plan and is subject to adjustment term by term. The instructor and the university reserve the right to modify elements of the course during the term. The university may change the dates and

deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.

Sprint	Design Steps	Duration
Sprint 1	Evaluate	2 weeks
Sprint 2	Conceptualize + Evaluate + Improve	3 weeks
Sprint 3	Ask & Define + Conceptualize + Evaluate + Improve	6 weeks

REQUIRED MATERIALS AND TEXTS

See Course Schedule for Assigned Readings available on Avenue.

CLASS FORMAT

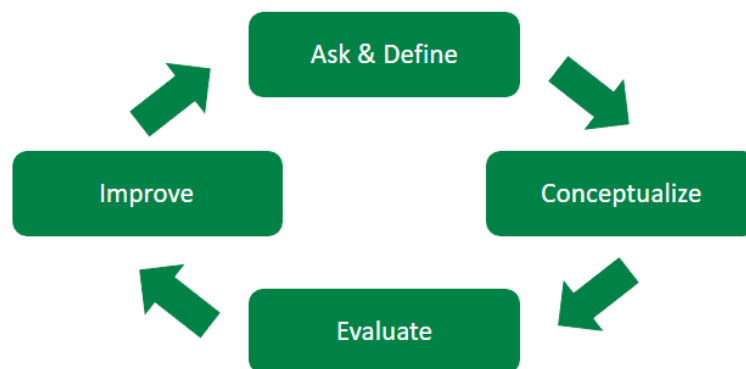
The course is scheduled for two in-person one-hour lectures per week, along with one in-person two-hour design studio per week.

Classes will consist of an interactive lecture format by the instructor, with students having the opportunity to ask questions concerning the information being provided to them.

The lectures will be recorded, and students are strongly encouraged to attend the in-person lectures.

In addition to formal class time, students are invited to meet with the instructor during regular office hours or by appointment to discuss course content or any other concerns.

Engineering design looks very different from one engineering discipline to another. For this course, the focus will be on the high-level design cycle below:



This course is delivered in three design sprints that focus on different steps of the design cycle. A design sprint, much like an athletic sprint, is designed to be a short-duration-high-intensity project. Each sprint in 2PX3 focuses on different steps of the design cycle. Each sprint includes a report, a peer evaluation, and a reflection that is submitted as the project concludes. Sprints 2 and 3 include a communication deliverable that assesses students'

ability to convey their design process using accessible language. Teams are shuffled between sprints (i.e., the teams that work together in Sprint 1 will not be the same teams working together in Sprint 2).

COURSE EVALUATION

Component	Weight (%)
DS Meeting 1 & DS Meeting 2 (2) **	10%
Individual Reflections (3)	10%
Individual Assignments (3)	10%
Final Presentation (Group)	7.5%
Final Presentation (Individual)	7.5%
Group Video	15%
Group Reports (3) *	40%
Total	100%
** 5% each, subject to attendance regrading scheme described below	
* Reports 1, 2, and 3 are weighted 10%, 10%, and 20%, respectively	

COURSE EVALUATION DETAILS

Attendance Reweight

Attendance will be taken during both lecture and design studio. For students that maintain an attendance rate of 75% or higher for both lectures and design studios, the weights of the assessments will be shifted to benefit students in the following ways:

- The lowest Individual Reflection grade will be dropped.
- The lowest Individual Assignment grade will be dropped.
- 2.5% of the weight of the lowest-graded Individual Presentation/Meeting will be shifted to the highest-graded Individual Presentation/Meeting.
- 5% of the weight from the lower grade between the Group Video and Individual Component of the Group Presentation will be shifted to the higher of the two (e.g., if your Group Video grade is higher than the Group Presentation, 5% of the weight from Group Presentation shifts to your Group Video grade).
- 5% of the weight from your lowest Group Report grade shifts to your highest Group Report grade.

Please note: any individually graded assessment that is dropped as a result of a McMaster Student Absence Form (MSAF) will count as the dropped assessment for the Attendance Reweight. For example, if a student meets the Attendance Reweight and uses an MSAF on one of the Reflections, the other two Reflections will count toward the student's final grade.

GRADING CONCERNS

Grades will be posted to Avenue as soon as possible upon completion. You will have 7 calendar days from the date your grade is posted to address any concerns you may have to the instructional team.

- Any questions/concerns must be addressed electronically through an MS Form that will be made available on Avenue.
- Concerns with regards to grading will not be considered without submitting through the appropriate channels (i.e., the posted MS Form).
- Any concerns or appeals brought to the attention of the instructional team after 7 calendar days from the day the grade was posted to Avenue will not be processed.

GRADING SCALE

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

CONDUCT EXPECTATIONS

As a McMaster graduate student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

Students must be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in university activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

TURNITIN.COM

Some courses may use a web-based service (Turnitin.com) to reveal the authenticity and ownership of student-submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., online search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

GENERATIVE AI: UNRESTRICTED USE

Students may use generative AI throughout this course in whatever way enhances their learning; no special documentation or citation is required.

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. The academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.** Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. a grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>.

The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

COURSES WITH AN ON-LINE ELEMENT

In this course, we will be using Avenue to Learn. Students should be aware that, when they access the electronic components of this course, private information such as first and last names, usernames for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation of this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

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EQUITY, DIVERSITY & INCLUSION

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, particularly if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or the [Equity and Inclusion Office](#).

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Assessments that are submitted as a group will not be reweighted to other assessments if an MSAF is used at the time they are due.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS, OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should normally submit their request to their Faculty Office within 10 working days of the beginning of the term in which they anticipate a need for accommodation or to the Registrar's Office before their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright-protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by university instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for authorized distribution, or by a student for personal study. Students should be aware that their voices and/or images may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.